TEAM: A3-PS021-TC450

PS-021: Experimental Dialog Correlation and Recovery Across Agent Adapters

Building a prototype for reliable task identity, duplicate suppression, and crash recovery.

6G & FUTURE NETWORKS · PYTHON · FASTAPI · SQLITE · HTTP/JSON

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CACHE US IF YOU CAN

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THE FAILURE BOUNDARY

Problem Description

The missing response is not proof of a missing execution

Different agent adapters can represent context and task lifecycle differently.

A timeout leaves the client uncertain: the server may have executed the request.

Retries without stable identity can repeat the same business operation.

A restart can interrupt work between acceptance, execution, and completion.

FAILURE SEQUENCE



CONTEXT

Background

Two adapters must agree on one task identity across retries and restarts.

Client adapter

Caller, envelope formatter, HTTP client, and retry manager can resend the same logical request.

Server adapter

FastAPI handler, idempotency engine, and SQLite ledger decide whether work already happened.

SCOPE: EXPERIMENTAL PROTOTYPE OF THE ADAPTER BOUNDARY, NOT A COMPLETE AGENT PROTOCOL IMPLEMENTATION.

STANDARDS CONTEXT

Literature & RFC Review

IETF Agent Protocol group

Working group context for agent protocol work: https://datatracker.ietf.org/group/agentproto/about/

Boundary of this review

Reviewed as context only. This prototype does not claim IETF compliance, complete MCP/A2A interoperability, or universal exactly-once delivery.

THE OFFICIAL STRUCTURE PDF SUPPLIED WITH THE REQUEST DEFINES THE SLIDE ORDER.

IDENTITY MODEL

Proposed Solution

One logical task, three identities, one retry counter, explicit state

dialog_id

Stable identity of the logical conversation or workflow.

idempotency_key

Stable identity of one intended business side effect across retries.

request_id

Unique identity of each transmission attempt.

attempt

Monotonically increasing retry count.

Attempt identity

FIELD	ATTEMPT 1	ATTEMPT 2
dialog_id	dlg_42	dlg_42
idempotency_key	idem_42	idem_42
request_id	req_A1	req_A2
attempt	1	2

Gold = changes on every retry. Everything else stays the same.

State path

RECEIVED → ACCEPTED → RUNNING → COMPLETED

Recovery path: RECOVERING → RUNNING → COMPLETED

Different payload under the same idempotency key becomes a conflict.

SAFE REGISTRATION

Optimization Proposed

Atomic registration turns retries into a safe lookup

unique key

SQLite uniqueness constraint on idempotency_key.

serialize

BEGIN IMMEDIATE serializes competing registration transactions.

register

No non-transactional check-then-insert race.

lookup

Completed duplicate returns the stored result; running duplicate returns HTTP 202.

CRITICAL BOUNDARY

Local side effect + COMPLETED transition

Commit in the same SQLite transaction.

CRASH OUTCOMES

Before commit: both roll back.

After commit: both persist.

Different payload: HTTP 409 Conflict.

Before: check-then-insert can register duplicates under concurrent retries. **After:** the unique key and BEGIN IMMEDIATE turn duplicates into a lookup.

THIS DOES NOT MAKE EXTERNAL PROVIDERS, THIRD-PARTY APIS, OR MODEL INFERENCE EXACTLY ONCE.

SYSTEM BOUNDARY

Solution Architecture & Design

Adapter A

caller · envelope formatter · HTTP client · retry manager

Transport

POST /dialog/task

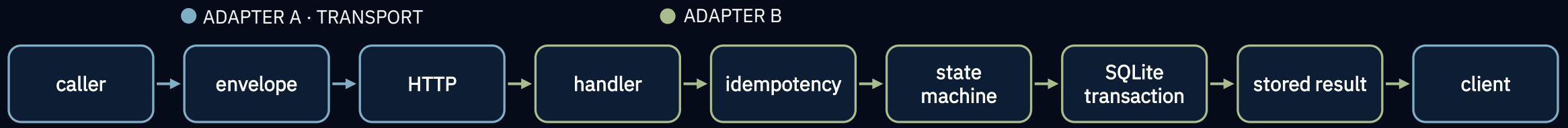
FRESH REQUEST_ID PER TRANSMISSION · 200 COMPLETED · 202 IN PROGRESS · 409 PAYLOAD CONFLICT

Adapter B

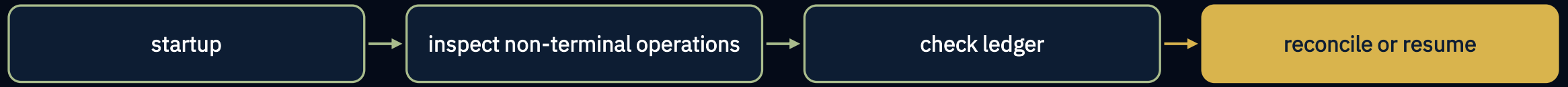
FastAPI handler · idempotency engine · atomic registration · lifecycle state machine · business logic · startup recovery

SQLite

dialogs · operations · side_effects ledger · event_log



RECOVERY FLOW



STARTUP RECOVERY

Data Flow Diagram

Recovery preserves identity, checks durable evidence, then chooses reconcile or resume.

ENTRY
FastAPI lifespan invokes StartupRecoveryManager.

SCOPE
Inspect RECEIVED, ACCEPTED, RUNNING, and RECOVERING operations.

EVIDENCE
Query the durable side_effects ledger.

GATE
Preserve dialog_id and idempotency_key; accept inbound traffic after recovery completes.

EFFECT COMMITTED

Reconcile

Reconcile stored details to COMPLETED.

EFFECT NOT COMMITTED

Resume

Enter RECOVERING and resume deterministic work safely; commit effect and COMPLETED atomically.

PHASES 1–4 · VALIDATION GATE

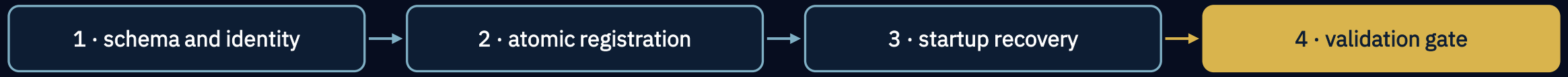
Deployment Timeline

Five failure scenarios make the boundary testable before promotion

Scenario	Expected behavior	Evidence to inspect
Lost response after completion	Retry returns stored result.	state · result · ledger count · HTTP response
Concurrent duplicate requests	One operation is created.	uniqueness · rows · event log
Crash before commit	Local transaction rolls back.	absent effect · recoverable state
Crash after commit	Recovery recognizes the effect and reconciles completion.	ledger · stored result
Mixed recovery + retry	Original dialog identity survives; committed effects stay suppressed.	identity · ledger · event log

NO DATES, TEST RESULTS, PERFORMANCE METRICS, OR SUCCESS PERCENTAGES ARE ASSERTED. EXPECTED BEHAVIOR AND EVIDENCE ARE DEFINED HERE.

PHASES



REFERENCES

Bibliography I

1. IETF Agent Protocol working group: https://datatracker.ietf.org/group/agentproto/about/
2. Official structure PDF supplied with this request (PS-021 report sections)
3. FastAPI documentation: https://fastapi.tiangolo.com/
4. SQLite documentation, transactions and BEGIN IMMEDIATE: https://www.sqlite.org/lang_transaction.html

PS-021 · PROTOTYPE BOUNDARY

Thank You for Your Attention!

Reliable recovery within a clearly defined boundary: no network-failure elimination, no universal exactly-once claim.

CORRELATION → DURABLE STATE → IDEMPOTENCY → RECOVERY → VALIDATION

EXTERNAL EFFECTS NEED PROVIDER-SIDE IDEMPOTENCY, AN OUTBOX, RECONCILIATION, OR COMPENSATING WORKFLOWS.

WHAT THE PROTOTYPE PRESERVES

Stable correlation across retries.

Durable operation state across restarts.

Transactional idempotency for duplicate local effects.

Deterministic recovery from committed database evidence.

BOUNDARY

Experimental schema. No complete MCP/A2A interoperability claim. No universal exactly-once delivery across uncoordinated boundaries.